

PRANVEER SINGH INSTITUTE OF TECHNOLOGY

B.Tech Computer Science and Engineering — Semester Examination Simulation

SUBJECT: WEB TECHNOLOGY

Paper Set:	Set 02	Duration:	3 Hours
Maximum Marks:	100 Marks	Date:	September 2026

SECTION A

Answer all questions in brief. (10 x 2 = 20 Marks)

- Q1(a).** Write down the basic properties or conditions required for Borders. [2M]
- Q1(b).** Differentiate in brief between HTML and CSS Selectors. [2M]
- Q1(c).** List any two advantages and limitations of CSS Selectors. [2M]
- Q1(d).** Give a real-world example illustrating XML. [2M]
- Q1(e).** Define CSS Box Model and state its key application in computer systems. [2M]
- Q1(f).** What is the primary significance of Margins? Explain in brief. [2M]
- Q1(g).** State the fundamental principles governing CSS for Lists. [2M]
- Q1(h).** Write down the basic properties or conditions required for Web Development. [2M]
- Q1(i).** Differentiate in brief between Types of CSS and Borders. [2M]
- Q1(j).** List any two advantages and limitations of Borders. [2M]

SECTION B

Attempt any three of the following questions. (3 x 10 = 30 Marks)

- Q2.** Design a solution using HTML for an enterprise system and trace each step. [10M]
- Q3.** Explain the theoretical foundation of CSS Selectors and prove its correctness. [10M]
- Q4.** Write a step-by-step algorithm or procedure for XML with illustrative inputs. [10M]
- Q5.** Discuss the various challenges encountered in CSS Box Model and how they are mitigated. [10M]
- Q6.** Explain Margins in detail with suitable architectural diagrams and examples. [10M]

SECTION C

Attempt any five questions from this section. (5 x 10 = 50 Marks)

- Q5.** Design a solution using CSS for Lists for an enterprise system and trace each step. [10M]
- Q6.** Explain the theoretical foundation of Web Development and prove its correctness. [10M]
- Q7.** Write a step-by-step algorithm or procedure for Types of CSS with illustrative inputs. [10M]
- Q8.** Discuss the various challenges encountered in Borders and how they are mitigated. [10M]
- Q9.** Explain HTML in detail with suitable architectural diagrams and examples. [10M]
- Q10.** Discuss the algorithmic approach for CSS Selectors. Analyze its time and space complexity. [10M]

Q11. Compare and contrast XML with CSS Box Model. Highlight performance trade-offs.

[10M]

— Generated by Precedent Academic Intelligence Engine —